SIMATS Online Programming Lab

PYTHON PROGRAMMING

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Q2 **Void Function (No Return Value)** 5 marks

Write a function to print the multiplication table of a given number.

Your Code

```
def multiplication_table(n):
    for i in range(1, 11):
        print(f"{n} x {i} = {n * i}")

n = int(input())
multiplication_table(n)
```

Input

```
5
```

Output

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
```

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Q3 **Void Function (No Return Value)** 5 marks

Write a function to print all even numbers from 1 to N.

Your Code

```
def print_even_numbers(n):
    for i in range(2, n + 1, 2):
        print(i)

n = int(input())
print_even_numbers(n)
```

Input

```
5
```

Output

```
2
4
```

Submitted

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Q4 **Fruitful Function (Returns One Value)** 5 marks

Write a function that accepts a string and returns the number of vowels in it.

Your Code

```
def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count
s = input()
print(count_vowels(s))
```

Input

```
A B
```

Output

```
1
```

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Q5 **Fruitful Function (Returns One Value)**

5 marks

Write a function that accepts a list of integers and returns the largest element.

Your Code

```python
def largest_element(numbers):
    largest = numbers[0]
    for i in numbers:
        if i > largest:
            largest = i
    return largest

# Input
numbers = list(map(int, input("Enter numbers separated by space: ").spl

# Function call
result = largest_element(numbers)

# Output
print("Largest element is:", result)
```

Input

```
10 25 8 45 18
45
```

Output

```
Enter numbers separated by space:
Largest element is: 45
```

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Q6 **Fruitful Function (Returns One Value)**

5 marks

Write a function that accepts a number and returns whether it is prime or not.

Your Code

```python
def is_prime(n):
    if n < 2:
        return False

    for i in range(2, n):
        if n % i == 0:
            return False

    return True

# Input
num = int(input("Enter a number: "))

# Output
if is_prime(num):
    print("Prime Number")
else:
    print("Not a Prime Number")
```

Input

```
13
```

Output

```
Enter a number: Prime Number
```

Q7 **Function Returning Two or More Values** 5 marks

Write a function that accepts a student's marks in three subjects and returns the total, average, and grade.

Your Code

```
def student_result(m1, m2, m3):
    total = m1 + m2 + m3
    average = total / 3

    if average >= 90:
        grade = "A"
    elif average >= 75:
        grade = "B"
    elif average >= 50:
        grade = "C"
    else:
        grade = "F"

    return total, average, grade

# Input
m1 = int(input("Enter marks in Subject 1: "))
m2 = int(input("Enter marks in Subject 2: "))
m3 = int(input("Enter marks in Subject 3: "))

# Function call
total, average, grade = student_result(m1, m2, m3)
```

Input

```
80
90
85
```

Output

```
Enter marks in Subject 1: Enter
marks in Subject 2: Enter marks
in Subject 3: Total = 255
Average = 85.0
Grade = B
```

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Q8 Function Returning Two or More Values

5 marks

Write a function that accepts the radius of a circle and returns its area and circumference.

Your Code

```python
def circle(radius):
    area = 3.14 * radius * radius
    circumference = 2 * 3.14 * radius
    return area, circumference

# Input
r = float(input("Enter radius: "))

# Function call
area, circumference = circle(r)

# Output
print("Area =", area)
print("Circumference =", circumference)
```

Input

```
5
78.5
31.5
```

Output

Run

Save

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Q9 Function Returning Two or More Values

5 marks

Write a function that accepts a number and returns its square, cube, and square root.

Your Code

```python
import math

def calculate(n):
    square = n ** 2
    cube = n ** 3
    square_root = math.sqrt(n)
    return square, cube, square_root

num = float(input("Enter a number: "))

sq, cb, root = calculate(num)

print("Square =", sq)
print("Cube =", cb)
print("Square Root =", root)
```

Input

```
9
```

Output

```
Enter a number: Square = 81.0
Cube = 729.0
Square Root = 3.0
```

Run Save

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Q10 **Keyword Arguments** 5 marks

Write a function student(name, age, department) and call it using keyword arguments.

Your Code

```python
def student(name, age, department):
    print("Name :", name)
    print("Age :", age)
    print("Department :", department)

# Function call using keyword arguments
student(name="Rahul", age=20, department="CSE")
```

Input

stdin input

Output

```
Name : Rahul
Age : 20
Department : CSE
```